

A confluent co-de-Bruijn σ -calculus: a signature and its term rewriting system

We present substitution for an intrinsically scoped, co-de-Bruijn object language as an abstract *symbolic rewrite system*: a typed signature of operator symbols (Part), and a set of oriented rewrite rules over those symbols (Part 6). The presentation is purely syntactic. Each rule group carries one tag:

- (L) **Library.** Object-language-independent; fixed once.
- (D) **Derivable.** Syntax-dependent but mechanical: generated from one traversal $[\cdot]$ per object syntax together with three base rules (FUSE, IDEMB, VARCONS).
- (S) **Syntax-specific.** Supplied per object language: typing and reduction.

Part I. Signature

Scopes $\Gamma, \Delta, \Upsilon, \dots$: Scope are lists of sorts; s ranges over sorts. Below, each symbol is declared with its arity/type only.

1 Thinnings (L)

A thinning $\theta : \Gamma \sqsubseteq \Delta$ relates two scopes.

$\text{oz} : [] \sqsubseteq []$	empty-scope thinning
$\text{os}(\cdot) : (\Gamma \sqsubseteq \Delta) \rightarrow (s :: \Gamma \sqsubseteq s :: \Delta)$	keep
$\text{o}'(\cdot) : (\Gamma \sqsubseteq \Delta) \rightarrow (\Gamma \sqsubseteq s :: \Delta)$	drop
$1 : \Gamma \sqsubseteq \Gamma$	identity
$\circ : (\Gamma \sqsubseteq \Delta) \rightarrow (\Delta \sqsubseteq \Upsilon) \rightarrow (\Gamma \sqsubseteq \Upsilon)$	composition
$\text{e} : [] \sqsubseteq \Delta$	empty thinning

2 Covers and coproducts (L)

A cover $c : \Gamma_l \mid \Gamma_r \blacktriangleright \Gamma$ is built from czz , css , cs' , c's (empty / shared / left-only / right-only).

$\text{thinL}(\cdot) : \Gamma_l \mid \Gamma_r \blacktriangleright \Gamma \rightarrow (\Gamma_l \sqsubseteq \Gamma)$	left embedding of a cover
$\text{thinR}(\cdot) : \Gamma_l \mid \Gamma_r \blacktriangleright \Gamma \rightarrow (\Gamma_r \sqsubseteq \Gamma)$	right embedding
$\sqcup : (\Gamma_1 \sqsubseteq \Delta) \rightarrow (\Gamma_2 \sqsubseteq \Delta) \rightarrow \text{Cop}$	coproduct of two thinnings
$\text{out}(\cdot) : \text{Cop} \rightarrow (\Upsilon \sqsubseteq \Delta)$	outer thinning of a coproduct
$\text{cov}(\cdot) : \text{Cop} \rightarrow \Gamma_1 \mid \Gamma_2 \blacktriangleright \Upsilon$	cover of a coproduct
$\text{mkCop} : (\Gamma_1 \sqsubseteq \Upsilon)(\Gamma_2 \sqsubseteq \Upsilon)(\Upsilon \sqsubseteq \Delta) \Gamma_1 \mid \Gamma_2 \blacktriangleright \Upsilon \rightarrow \text{Cop}$	coproduct constructor
$\text{covL}(\cdot) : (\Gamma \sqsubseteq \Delta) \rightarrow \Delta \mid \Gamma \blacktriangleright \Delta$	left-canonical cover
$\text{covR}(\cdot) : (\Gamma \sqsubseteq \Delta) \rightarrow \Gamma \mid \Delta \blacktriangleright \Delta$	right-canonical cover
$\text{full} : \Gamma \mid \Gamma \blacktriangleright \Gamma$	diagonal cover

The fields of a `mkCop` value are read off by `out` and `cov`: `out(mkCop l r o c) = o` and `cov(mkCop l r o c) = c`.

3 Things-with-thinnings (L)

For a scope-indexed family T , the type $T\uparrow\Delta$ pairs a term over some support Υ with a thinning $\Upsilon \sqsubseteq \Delta$; the pair is written $t\uparrow\theta$.

$\cdot\uparrow\cdot : T\Upsilon \rightarrow (\Upsilon \sqsubseteq \Delta) \rightarrow T\uparrow\Delta$	pairing
$\cdot\langle\cdot\rangle : T\uparrow\Gamma \rightarrow (\Gamma \sqsubseteq \Delta) \rightarrow T\uparrow\Delta$	renaming
$\langle\$ \rangle : (T\Gamma \rightarrow U\Gamma) \rightarrow T\uparrow\Delta \rightarrow U\uparrow\Delta$	map under a thinning ($f\langle\$ \rangle u$)
$\times^r : (T\Gamma)(U\Gamma) \rightarrow (T \times^r U)\Delta$	relevant pairing
<code>Bind bT : Scope $\rightarrow$ Set</code>	a T under one bound variable of sort b

The *relevant pairing* $u \times^r v : (T \times^r U)\Delta$ merges two things-with-thinnings $u : T\uparrow\Delta$, $v : U\uparrow\Delta$ into one node carrying the cover $\text{cov}(\theta \sqcup \phi)$ of their two supports and the outer thinning $\text{out}(\theta \sqcup \phi)$. The binder former `Bind  $bT$` packages a T together with a thinning recording whether the bound b -variable is used or dropped. The map $\langle\$ \rangle$ relabels the term inside a thing-with-thinning, leaving its thinning fixed.

4 Substitutions (L)

A substitution $\sigma : \Gamma \Rightarrow \Delta$ carries one $\cdot\uparrow\Delta$ per variable of Γ .

$\text{id} : \Gamma \Rightarrow \Gamma$	identity
$\uparrow : \Gamma \Rightarrow (s :: \Gamma)$	shift
$[\cdot] : T\Gamma \rightarrow \Gamma \Rightarrow \Delta \rightarrow T\uparrow\Delta$	instantiation $t[\sigma]$
$\cdot\llbracket\cdot\rrbracket : T\uparrow\Gamma \rightarrow \Gamma \Rightarrow \Delta \rightarrow T\uparrow\Delta$	apply to a thing $u\llbracket\tau\rrbracket$
$\circ : \Gamma \Rightarrow \Delta \rightarrow \Delta \Rightarrow \Upsilon \rightarrow \Gamma \Rightarrow \Upsilon$	composition $\sigma \circ \tau$
$\bullet : T\uparrow\Delta \rightarrow \Gamma \Rightarrow \Delta \rightarrow (s :: \Gamma) \Rightarrow \Delta$	cons $u \bullet \sigma$
$\text{v}_0 : T\uparrow(s :: \Delta)$	first variable (= <code>var↑ose</code>)
$\langle \rangle : [] \Rightarrow \Delta$	empty substitution
$\text{emb}(\cdot) : (\Gamma \sqsubseteq \Delta) \rightarrow \Gamma \Rightarrow \Delta$	embedding of a thinning
$\downarrow_L, \downarrow_R : \Gamma_l \mid \Gamma_r \blacktriangleright \Gamma \rightarrow \Gamma \Rightarrow \Delta \rightarrow \Gamma_l/\Gamma_r \Rightarrow \Delta$	cover split
$\upharpoonright : \Delta \Rightarrow \Upsilon \rightarrow (\Gamma \sqsubseteq \Delta) \rightarrow \Gamma \Rightarrow \Upsilon$	restriction $\sigma \upharpoonright \theta$
$\text{wk}(\cdot) : \Gamma \Rightarrow \Delta \rightarrow \Gamma \Rightarrow (s :: \Delta)$	weakening

Defined abbreviation (binder lift).

$$\uparrow(\cdot) : \Gamma \Rightarrow \Delta \rightarrow (s :: \Gamma) \Rightarrow (s :: \Delta), \quad \uparrow\sigma := \text{v}_0 \bullet (\sigma \circ \uparrow).$$

The shift $\uparrow$ is the primitive; general weakening is the defined $\text{wk}\sigma := \sigma \circ \uparrow$.

5 Contexts (L)

A context $\Phi : \text{Cx}\Gamma$ carries one closed classifier per variable of Γ .

$\varepsilon : \text{Cx}[]$	empty context
$\cdot, \cdot : \text{Cx}\Gamma \rightarrow \text{Ent} \rightarrow \text{Cx}(s :: \Gamma)$	context extension
$\upharpoonright : \text{Cx}\Delta \rightarrow (\Gamma \sqsubseteq \Delta) \rightarrow \text{Cx}\Gamma$	restriction $\Phi \upharpoonright \theta$
$\text{splitL}, \text{splitR} : \Gamma_l \mid \Gamma_r \blacktriangleright \Gamma \rightarrow \text{Cx}\Gamma \rightarrow \text{Cx}\Gamma_{l/r}$	cover split

The split is defined: $\text{splitL } c \Phi := \Phi \upharpoonright (\text{thinL } c)$, $\text{splitR } c \Phi := \Phi \upharpoonright (\text{thinR } c)$.

6 Object syntax (S)

STLC. A single family $\text{Tm} : \text{Scope} \rightarrow \text{Set}$:

$$\begin{array}{ll} \text{var} : \text{Tm} (s :: []) & \\ \text{app} : (\text{Tm} \times^r \text{Tm}) \Gamma \rightarrow \text{Tm} \Gamma & \text{stores a cover of the two operand supports} \\ \text{lam} : \text{Bind Tm } \Gamma \rightarrow \text{Tm } \Gamma & \text{binder body: use or drop} \end{array}$$

Smart constructors $\text{app}^\uparrow, \lambda^\uparrow$ pair via the coproduct and bind via the drop-thinning.

System F. A single sorted family $\text{Exp} : \text{Scope} \rightarrow \text{Sort} \rightarrow \text{Set}$ with $\text{Sort} = \{\text{ty}, \text{tm}\}$, writing $\text{Ty} = \text{Exp}^{\text{ty}}$, $\text{Tm} = \text{Exp}^{\text{tm}}$:

$$\begin{array}{lll} \text{var} : \text{Exp}^r (r :: []) & _ \dashv _ : (\text{Ty} \times^r \text{Ty}) \Gamma \rightarrow \text{Ty} \Gamma & \forall : \text{Bind ty Ty } \Gamma \rightarrow \text{Ty} \Gamma \\ \text{'lam} : (\text{Ty} \times^r \text{Bind tm Tm}) \Gamma \rightarrow \text{Tm} \Gamma & \text{'Lam} : \text{Bind ty Tm } \Gamma \rightarrow \text{Tm} \Gamma & \text{'app} : (\text{Tm} \times^r \text{Tm}) \Gamma \rightarrow \text{Tm} \Gamma \\ \text{'App} : (\text{Tm} \times^r \text{Ty}) \Gamma \rightarrow \text{Tm} \Gamma & & \end{array}$$

A single substitution $[\cdot]$ acts at both sorts; the lift under every binder ($\forall, \text{'lam}, \text{'Lam}$) is the same $\uparrow$.

Part II. Rewrite rules

Every rule is oriented left-to-right. We write $\text{name} : \text{LHS} \rightarrow \text{RHS}$.

7 (R1) Thinning monoid (L)

$$\text{IDL}_{\sqsubseteq} : 1 \circ \theta \rightarrow \theta \tag{1}$$

$$\text{IDR}_{\sqsubseteq} : \theta \circ 1 \rightarrow \theta \tag{2}$$

$$\text{ASSOC}_{\sqsubseteq} : (\theta \circ \phi) \circ \psi \rightarrow \theta \circ (\phi \circ \psi) \tag{3}$$

8 (R2) Coproduct / cover algebra (L)

In each slice $\sqsubseteq / \Delta$, two subsopes $\theta : \Gamma_1 \sqsubseteq \Delta$, $\phi : \Gamma_2 \sqsubseteq \Delta$ have a *union* — their coproduct. The symbol $\sqcup$ computes it: $\text{out}(\theta \sqcup \phi)$ is its inclusion into Δ and $\text{thinL}(\text{cov}(\theta \sqcup \phi)), \text{thinR}(\text{cov}(\theta \sqcup \phi))$ are its two injections, with $\text{thinL}(\text{cov}(\theta \sqcup \phi)) \circ \text{out}(\theta \sqcup \phi) = \theta$ and symmetrically on the right. The canonical covers are the degenerate coproducts: $\text{covL } \phi = \phi \sqcup 1$, $\text{covR } \theta = 1 \sqcup \theta$, $\text{full} = 1 \sqcup 1$. Group (R2) is exactly the β /unit laws of this coproduct.

$$\text{COP-UNITL} : 1 \sqcup \phi \rightarrow \text{mkCop } 1 \phi 1 (\text{covL } \phi) \tag{4}$$

$$\text{COP-UNITR} : \theta \sqcup 1 \rightarrow \text{mkCop } \theta 1 1 (\text{covR } \theta) \tag{5}$$

$$\text{COVL-ID} : \text{covL } 1 \rightarrow \text{full} \tag{6}$$

$$\text{COVR-ID} : \text{covR } 1 \rightarrow \text{full} \tag{7}$$

$$\text{THINL-COVL} : \text{thinL}(\text{covL } \phi) \rightarrow 1 \tag{8}$$

$$\text{THINR-COVL} : \text{thinR}(\text{covL } \phi) \rightarrow \phi \tag{8}$$

$$\text{THINL-COVR} : \text{thinL}(\text{covR } \theta) \rightarrow \theta \tag{9}$$

$$\text{THINR-COVR} : \text{thinR}(\text{covR } \theta) \rightarrow 1 \tag{9}$$

$$\text{THINL-FULL} : \text{thinL } \text{full} \rightarrow 1 \tag{10}$$

$$\text{THINR-FULL} : \text{thinR } \text{full} \rightarrow 1 \tag{10}$$

$$\begin{array}{ll} \text{FAC-L} : \text{thinL}(\text{cov}(\theta \sqcup \phi)) \circ \text{out}(\theta \sqcup \phi) \rightarrow \theta & \text{FAC-R} : \text{thinR}(\text{cov}(\theta \sqcup \phi)) \circ \text{out}(\theta \sqcup \phi) \rightarrow \phi \end{array} \tag{11}$$

FAC-L/FAC-R are the injection factorisations: composing an injection with the inclusion recovers the original subscope.

9 (R3) σ_{SP} — instantiation (D)

c denotes the cover stored in the relevant **app**/pair node.

$$\text{VARCONS} : \text{var}[u \bullet \sigma] \longrightarrow u \quad (12)$$

$$\text{INST-APP} : (l \ r)[\sigma] \longrightarrow (l[c|_L \sigma]) (r[c|_R \sigma]) \quad (13)$$

$$\text{INST-LAM} : (\lambda^{\text{use}} t)[\sigma] \longrightarrow \lambda^\uparrow (t[\uparrow \sigma]) \quad (14)$$

$$\text{INST-LAM-DROP} : (\lambda^{\text{drop}} t)[\sigma] \longrightarrow \lambda^\uparrow (\text{drop } \langle \$ \rangle (t[\sigma])) \quad (15)$$

$$\text{IDSUBST} : t[\text{id}] \longrightarrow t\uparrow 1 \quad (16)$$

$$\text{CLOS} : (u[\![\tau]\!])[\![v]\!] \longrightarrow u[\![\tau \circ v]\!]$$

The instantiation rules **VARCONS**, **INST-APP**, **INST-LAM**, **INST-LAM-DROP** are the clauses of $[\cdot]$ on each constructor.

10 (R4) σ_{SP} — composition monoid (D)

$$\text{MAP} : (u \bullet \sigma) \circ \tau \longrightarrow (u[\![\tau]\!]) \bullet (\sigma \circ \tau) \quad (18)$$

$$\text{EMPTYCOMP} : \langle \rangle \circ \tau \longrightarrow \langle \rangle \quad (19)$$

$$\text{IDL} : \text{id} \circ \sigma \longrightarrow \sigma \quad (20)$$

$$\text{IDR} : \sigma \circ \text{id} \longrightarrow \sigma \quad (21)$$

$$\text{ASSOC} : (\sigma \circ \tau) \circ v \longrightarrow \sigma \circ (\tau \circ v) \quad (22)$$

11 (R5) σ_{SP} — **cons** / η (D)

$$\text{SHIFTCONS} : \uparrow \circ (u \bullet \sigma) \longrightarrow \sigma \quad (23)$$

$$\text{IDCONS} : v_0 \bullet \uparrow \longrightarrow \text{id} \quad (24)$$

$$\text{SCONS} : (v_0[\sigma]) \bullet (\uparrow \circ \sigma) \longrightarrow \sigma \quad (25)$$

SCONS is the fundamental **cons**/ η law (surjective pairing of substitutions): every substitution out of $s :: \Gamma$ is its head $v_0[\sigma]$ consed onto its tail $\uparrow \circ \sigma$. **SHIFTCONS** is its tail projection and **IDCONS** its $\sigma = \text{id}$ instance; both are derivable from **SCONS** together with **VARCONS**. All three are **(D)**; they are registered together only because the confluent completion needs them all present.

Derivability of (R4)/(R5). The whole of groups (R4) and (R5) is generated from three base rules together with the clauses of $[\cdot]$:

$$\text{FUSE} : (t[\sigma])[\![v]\!] \longrightarrow t[(\sigma \circ v)], \quad \text{IDEMB} : t[\text{emb } \theta] \longrightarrow t\uparrow \theta, \quad \text{VARCONS} : \text{var}[u \bullet \langle \rangle] \longrightarrow u.$$

12 (R6) Context coherence (L)

Restriction makes **Cx** a *presheaf* on the thinning category: **REST-ID** and **REST-FUN** are its functoriality, and **REST-EMPTY** fixes its value at the empty thinning **e**.

$$\text{REST-ID} : \Phi \upharpoonright 1 \longrightarrow \Phi \quad (26)$$

$$\text{REST-EMPTY} : \Phi \upharpoonright \mathbf{e} \longrightarrow \varepsilon \quad (27)$$

$$\text{REST-FUN} : \Phi \upharpoonright (\psi \circ \theta) \longrightarrow (\Phi \upharpoonright \theta) \upharpoonright \psi \quad (28)$$

$$\text{COHL} : (\Phi \upharpoonright \text{out}(\theta \sqcup \phi)) \upharpoonright \text{thinL}(\text{cov}(\theta \sqcup \phi)) \longrightarrow \Phi \upharpoonright \theta \quad (29)$$

$$\text{COHR} : (\Phi \upharpoonright \text{out}(\theta \sqcup \phi)) \upharpoonright \text{thinR}(\text{cov}(\theta \sqcup \phi)) \longrightarrow \Phi \upharpoonright \phi \quad (30)$$

CohL/CohR are derived. The coherences are not primitive: COHL follows from functoriality (REST-FUN) and the thinning-level factorisation FAC-L (Eq. 11):

$$(\Phi \upharpoonright \text{out}(\theta \sqcup \phi)) \upharpoonright \text{thinL}(\text{cov}(\theta \sqcup \phi)) \stackrel{\text{REST-FUN}}{=} \Phi \upharpoonright (\text{thinL}(\text{cov}(\theta \sqcup \phi)) \circ \text{out}(\theta \sqcup \phi)) \stackrel{\text{FAC-L}}{=} \Phi \upharpoonright \theta,$$

and COHR symmetrically from FAC-R. They are registered as rewrites in their own right only so that the rule set completes to a confluent system; mathematically they are theorems.

Meta-property

Proposition 1 (Local confluence). *The rule set (R1)–(R6) is locally confluent. Hence the induced equational theory holds definitionally: each rule may be used as a silent computation step.*

13 (R7) Typing and call-by-value reduction (S)

Typing $\Phi \vdash t : A$ is an extrinsic judgement over Cx , with $\Phi \vdash_{\uparrow} t \upharpoonright \theta : A \iff \Phi \upharpoonright \theta \vdash t : A$. For STLC:

$$\frac{}{(\varepsilon, A) \vdash \text{var} : A} \quad \frac{\Phi \upharpoonright \theta \vdash l : A \rightarrow B \quad \Phi \upharpoonright \phi \vdash r : A}{\Phi \upharpoonright \text{out}(\theta \sqcup \phi) \vdash \text{app}(l, r, \text{cov}(\theta \sqcup \phi)) : B} \quad \frac{(\Phi, A) \vdash t : B}{\Phi \vdash \lambda^{\text{use}} t : A \rightarrow B} \quad \frac{\Phi \vdash t : B}{\Phi \vdash \lambda^{\text{drop}} t : A \rightarrow B}$$

Values. $\text{Value}(\lambda^{\text{use}} t)$ and $\text{Value}(\lambda^{\text{drop}} t)$.

Reduction rules (c the cover of the **app** node; a contractum is a thing-with-thinning, its support possibly smaller):

$$\beta_{\text{use}} : (\lambda^{\text{use}} b) v \longrightarrow b[v \bullet \text{emb}(\text{thinL } c)] \quad \text{Value } v \quad (31)$$

$$\beta_{\text{drop}} : (\lambda^{\text{drop}} b) v \longrightarrow b[\text{emb}(\text{thinL } c)] \quad \text{Value } v \quad (32)$$

$$\xi\text{-FUN} : \text{app}^{\uparrow}(l, r) \longrightarrow \text{app}^{\uparrow}(l' \langle \text{thinL } c \rangle) (r \upharpoonright \text{thinR } c) \quad l \longrightarrow l' \quad (33)$$

$$\xi\text{-ARG} : \text{app}^{\uparrow}(v, r) \longrightarrow \text{app}^{\uparrow}(v \upharpoonright \text{thinL } c) (r' \langle \text{thinR } c \rangle) \quad \text{Value } v, r \longrightarrow r' \quad (34)$$

Substitution preserves typing. For a well-typed substitution σ (each entry typed in Ψ at the classifier Φ records),

$$\text{SUB-PRES} : \quad \Phi \vdash t : A \implies \Psi \vdash_{\uparrow} t[\sigma] : A.$$

Subject reduction.

$$\text{PRESERVE} : \quad \Phi \vdash t : A \wedge t \longrightarrow t' \implies \Phi \vdash_{\uparrow} t' : A.$$

The β -cases are SUB-PRES at the β -environment; the ξ -cases use $\text{app}^{\uparrow}$ together with REST-FUN to re-embed the reduced subterm. Typing and subject reduction for System F are not yet formalised (the substitution engine and rules (R1)–(R6) are complete for it).

Summary

Group	Tag	Rules
(R1) Thinning monoid	(L)	$\text{IDL}_{\sqsubseteq}, \text{IDR}_{\sqsubseteq}, \text{ASSOC}_{\sqsubseteq}$
(R2) Coproduct / cover algebra	(L)	$\text{COP-UNITL/R}, \text{COVL/R-ID}, \text{THINL/R-COVL/R-FULL}, \text{FAC-L/R}$
(R3) σ_{SP} instantiation	(D)	$\text{VARCONS}, \text{INST-APP/LAM/LAM-DROP}, \text{IDSUBST}, \text{CLOS}$
(R4) σ_{SP} monoid	(D)	$\text{MAP}, \text{EMPTYCOMP}, \text{IDL}, \text{IDR}, \text{ASSOC}$
(R5) σ_{SP} cons/ η	(D)	$\text{SHIFTCONS}, \text{IDCONS}, \text{SCONS}$
(R6) Context coherence	(L)	$\text{REST-ID}, \text{REST-EMPTY}, \text{REST-FUN}, \text{COHL}, \text{COHR}$
(R7) Typing / CBV	(S)	$\beta_{\text{use}}, \beta_{\text{drop}}, \xi\text{-fun}, \xi\text{-arg}; \text{SUB-PRES}, \text{PRESERVE}$